

Parameter-Efficient Dense Prediction for Cross-View Geo-Localisation in Dense Urban Environments

Lucas Scott, Sonya Coleman, Dermot Kerr, Philip Vance, Yunzhou Zhang

School of Computing, Engineering and Intelligent Systems, Ulster University, Northern Ireland

Abstract

Cross-view geo-localisation (CVGL) aims to estimate the geographic position of a ground-level query image by matching it against overhead reference imagery. Despite recent advancements in transformer-based architectures, robust matching in dense urban environments remains impeded by substantial viewpoint variations and structural ambiguities. Recent frameworks, such as CVSegGuide, have integrated semantic segmentation maps to provide high-level scene context and reduce this domain gap. This study evaluates the efficacy of such semantic components by isolating the segmentation task from the broader CVGL pipeline. We perform a rigorous comparison between the established Multi-Scale Transformer (MST) baseline and several U-Net family architectures. Using the Brooklyn and Queens dataset, we evaluate locally trained MST, U-Net, Lite U-Net, and U-Net++ models. Quantitative results demonstrate that convolutional encoder-decoder architectures consistently provide superior land-cover masks compared with the isolated transformer baseline. Specifically, the 5-level U-Net achieves a peak mean Intersection over Union ($mIoU$) of 74.17%, while the 4-level Lite U-Net demonstrates exceptional parameter efficiency, reaching 71.58% $mIoU$ with only 4.662M parameters. These findings suggest that dedicated segmentation architectures offer more reliable semantic guidance than current unified transformer branches, providing a robust foundation for future attention-guided CVGL systems.

Keywords: Cross-View Geo-Localisation, Semantic Segmentation, Parameter Efficiency, Dense Prediction, Urban Land-Cover Mapping.

1 Introduction

Cross-View Geo-Localisation (CVGL) facilitates autonomous navigation by aligning ground-level imagery with georeferenced satellite or aerial maps. While traditional global navigation satellite systems (GNSS) are susceptible to signal denial and jamming, visual localisation provides a passive and resilient alternative [Yue et al., 2025, Yao et al., 2024]. However, the drastic disparity in perspectives between street-level panoramas and overhead images creates a significant “domain gap”, complicating direct feature matching in complex urban scenes. Historically, models relied on low-level visual features; however, recent shifts toward high-level semantic understanding suggest that structural landmarks provide more robust cues for localisation than transient visual textures [Castaldo et al., 2015].

To mitigate the domain gap, researchers have increasingly utilised semantic segmentation maps as a bridge between ground and overhead perspectives. By describing scene information through categories such as buildings, roads, and vegetation, segmentation-guided systems can focus on permanent spatial configurations while ignoring transient environmental changes. The CVSegGuide framework [Xavier et al., 2025] exemplifies this approach, using land-cover maps to guide transformer-based attention. However, the performance of these guidance mechanisms is strictly upper-bounded by the fidelity of the segmentation component. If the underlying segmentation branch fails to accurately delineate building footprints or road boundaries, the resulting attention maps will introduce spatial ambiguity into the CVGL pipeline.

This dependency necessitates a modular evaluation of the segmentation component. While integrated CVGL models often optimise for retrieval or coordinate regression, the internal segmentation branch must be rigorously assessed as an independent dense prediction task to ensure maximum reliability. By prioritising the development and deployment of high-accuracy segmentation methods, we can provide future CVGL systems with the most precise semantic priors possible.

Within semantic segmentation, the U-Net architecture has remained a foundational framework due to its encoder-decoder structure and skip connections, which preserve fine-grained spatial information while progressively learning higher-level representations [Ronneberger et al., 2015]. These properties make U-Net particularly well suited to aerial land-cover mapping, where accurate boundary delineation between roads, buildings, vegetation and background regions is essential. Although transformer-based architectures have recently gained prominence for their ability to model long-range contextual relationships [Zheng et al., 2021], convolutional U-Net variants continue to offer strong localisation performance, architectural simplicity, and favourable computational efficiency. This makes the U-Net family a relevant and necessary benchmark when evaluating segmentation backbones for CVGL-oriented semantic priors.

This paper benchmarks the Multi-Scale Transformer (MST) backbone against established convolutional encoder-decoder architectures to identify the most effective framework for aerial land-cover mapping. We investigate the U-Net family, specifically analysing standard U-Net, Lite U-Net, and U-Net++, to determine if the local inductive biases of convolutions offer advantages over the global attention mechanisms used in current transformer-based branches. Our primary contributions include: (i) a performance comparison of transformer versus convolutional architectures for urban segmentation, (ii) a data-lineage study using the Brooklyn and Queens dataset, and (iii) an analysis of parameter efficiency across varying model depths.

The remainder of this paper is organised as follows: Section 2 details the dataset and class-reduction strategy; Section 3 describes the experimental methodology and model configurations; Section 4 presents and analyses the quantitative and qualitative results; and Section 5 discusses the implications for future integrated CVGL systems.

2 Dataset

The experiments utilise the Brooklyn and Queens (Near/Remote) dataset, which provides paired overhead and ground-level imagery for dense urban environments. Each overhead image is 256×256 pixels with a ground sampling distance (GSD) of approximately 30 cm [Workman et al., 2017, Workman et al., 2022]. To ensure a controlled comparison with the benchmarks established in CVSegGuide [Xavier et al., 2025], we filter the dataset to include only aerial images with at least one corresponding ground-level panorama. This yields a training set of 34,876 images from the Brooklyn subset and a validation set of 7,414 images from the Queens subset.



Figure 1: Example images from the Brooklyn (left two) and Queens (right two) subsets of the dataset.

While the original masks provide eight semantic classes, we retain the four primary categories: *Tree Canopy*, *Buildings*, *Roads*, and *Other Impervious*. This consolidation serves a dual purpose: it ensures di-

rect comparative alignment with the baseline framework and mitigates the class-imbalance issues caused by the extreme sparsity of categories such as *Water*, *Bare Soil*, and *Railroads* in the study area. Representative samples from the filtered dataset are shown in Figure 1, as visually evidenced in these examples, the four classes are well-represented across diverse urban scenes, capturing a balanced mix of structural features ranging from dense residential areas to road networks.

Pixels belonging to these omitted classes are excluded from the loss calculation during training and masked during evaluation. As demonstrated in Table 1, the resulting class distribution remains highly consistent between the training and validation subsets, ensuring that our performance metrics reflect the structurally significant urban features necessary for stable cross-view matching.

Table 1: Pixel distribution across Brooklyn (Training) and Queens (Validation) subsets.

Class	Brooklyn (Training)		Queens (Validation)	
	Pixels (M)	%	Pixels (M)	%
Tree Canopy	396.8	18.26	90.9	20.42
Buildings	708.3	32.59	125.6	28.21
Roads	399.3	18.37	94.5	21.24
Other Impervious	668.7	30.77	134.1	30.14

3 Methodology

To evaluate the impact of architectural selection on semantic guidance, we employ a two-stage experimental design: a baseline reconstruction of the CVSegGuide MST model [Xavier et al., 2025] followed by a controlled comparison against several convolutional alternatives. This modular approach allows for the independent assessment of segmentation quality before investigating its downstream effects on localisation.

3.1 Architectural Configurations

The primary comparison evaluates the Multi-Scale Transformer (MST) against three variants of the U-Net family. While the MST baseline evaluates the original framework’s outputs, our locally trained configurations use strictly aerial imagery to isolate architectural performance.

- **Local MST:** A transformer-based backbone that extracts feature representations at multiple spatial resolutions, upsampled to produce four-channel land-cover maps [Fan et al., 2021].
- **U-Net:** A convolutional encoder-decoder architecture utilising skip connections to preserve spatial detail across downsampling layers [Ronneberger et al., 2015].
- **Lite U-Net:** A low-complexity variant designed to assess the viability of lightweight architectures in resource-constrained CVGL applications [Li et al., 2024].
- **U-Net++:** A nested encoder-decoder structure with dense skip pathways designed to refine feature fusion between the encoder and decoder [Zhou et al., 2018].

For each U-Net variant, we implement both **4-level** and **5-level** depths. The 5-level configurations introduce an additional downsampling stage, effectively expanding the receptive field and parameter capacity to determine if performance gains result from structural innovation or simple model scale.

3.2 Training and Evaluation Protocol

All models are trained from scratch using randomly initialised weights and follow a standardised training protocol using 256×256 aerial inputs. In alignment with the baseline replication [Xavier et al., 2025], no RGB normalisation or data augmentation is applied. Training is conducted on the Brooklyn subset using the Adam optimiser ($lr = 1 \times 10^{-5}$) and a batch size of 5 for a maximum of 100 epochs. Early stopping is triggered after five epochs without improvement in validation micro Intersection over Union (IoU). The optimisation objective weights Binary Cross-Entropy (BCE) and Dice loss equally to balance pixel-level classification with region-level segmentation quality [Mao et al., 2023, Sudre et al., 2017].

Performance is primarily measured using the macro-averaged Mean IoU ($mIoU$) across the four retained classes ($C = 4$). For a specific class c , the metric is computed via the confusion-matrix formulation:

$$IoU_c = \frac{TP_c}{TP_c + FP_c + FN_c}, \quad mIoU = \frac{1}{C} \sum_{c=1}^C IoU_c \quad (1)$$

where TP_c , FP_c , and FN_c represent true positives, false positives, and false negatives, respectively. To ensure a robust assessment, these components are accumulated across the entire Queens validation set prior to calculation, mitigating the impact of image-level variance.

While $mIoU$ serves as the definitive metric for model comparison, the micro IoU formulation is utilised for training-time checkpointing. By aggregating counts across all pixels regardless of class, micro IoU provides a stable optimisation signal:

$$IoU_{\text{micro}} = \frac{\sum TP_c}{\sum TP_c + \sum FP_c + \sum FN_c} \quad (2)$$

This dual-metric approach ensures that optimisation reflects total pixel accuracy while final evaluations respect the equal importance of each environmental landmark. Training and inference were performed on a GTX 1080 GPU with 32 GB of RAM.

4 Results and Discussion

The performance analysis presented in Table 2 reveals a substantial accuracy gap between the Multi-Scale Transformer (MST) and the U-Net family. This section discusses the quantitative trade-offs in parameter efficiency and architectural depth, followed by a qualitative examination of segmentation consistency.

Table 2: Performance comparison using the Queens validation set and per-class IoU and macro mIoU.

Model	Level	Params	Per-Class IoU (%)				mIoU
			Tree	Bldgs	Roads	Other	
MST	—	55.45M	58.04	74.50	72.98	52.49	64.50
U-Net	4	36.96M	68.80	81.03	82.14	60.41	73.09
U-Net	5	148.12M	69.63	83.01	83.76	60.30	74.17
Lite U-Net	4	4.66M	68.04	80.76	79.60	57.94	71.58
Lite U-Net	5	8.28M	66.47	79.43	78.48	57.69	70.52
U-Net++	4	36.62M	68.02	82.58	82.99	60.13	73.43
U-Net++	5	147.67M	69.65	82.56	83.09	60.03	73.83

4.1 Quantitative Performance and Parameter Efficiency

The U-Net family consistently outperformed the MST across all evaluation metrics. Notably, the standard 4-level U-Net achieved a 13.3% relative improvement in $mIoU$ over the MST baseline while utilising 33% fewer

parameters. This disparity suggests that the local inductive biases of convolutional structures are more effective for capturing the precise geometric boundaries of urban landmarks than the global attention mechanisms of the MST.

Increasing architectural depth from Level 4 to Level 5 yielded marginal gains for standard and nested architectures. For instance, the Level 5 U-Net achieved the highest overall accuracy (74.17%), but required a four-fold increase in parameters (111M additional weights) to secure a 1.08% improvement. Conversely, the Lite U-Net family showed an inverse relationship at increased depths, with performance decreasing from 71.58% to 70.52%. This suggests that for lightweight models, excessive depth may lead to optimisation difficulties or over-parameterisation relative to the feature complexity of aerial imagery.

The Lite U-Net (Level 4) stands out as the most efficient configuration, outperforming the MST by 7% *mIoU* while being approximately 12 times smaller. This makes it a compelling candidate for real-time CVGL applications where inference latency and memory overhead are critical constraints.

4.2 Qualitative Analysis

Visual inspection of the segmentation masks (Figure 2) confirms the quantitative findings. The MST outputs exhibit significant fragmented noise and a lack of structural coherence, particularly at the edges of buildings. These “salt-and-pepper” artifacts would likely introduce ambiguity if used to guide a downstream localisation model.

In contrast, the U-Net architectures produce much cleaner, contiguous segments. The standard U-Net and U-Net++ excel at delineating sharp transitions between *Buildings* (red) and *Roads* (grey), preserving the rectangular geometry of urban blocks. While the 5-level models capture finer details through their expanded receptive fields, the 4-level variants already provide the “homography-like” stability necessary for effective cross-view matching. These results indicate that dedicated convolutional encoder-decoders are superior for providing the reliable semantic priors required for high-accuracy localisation.

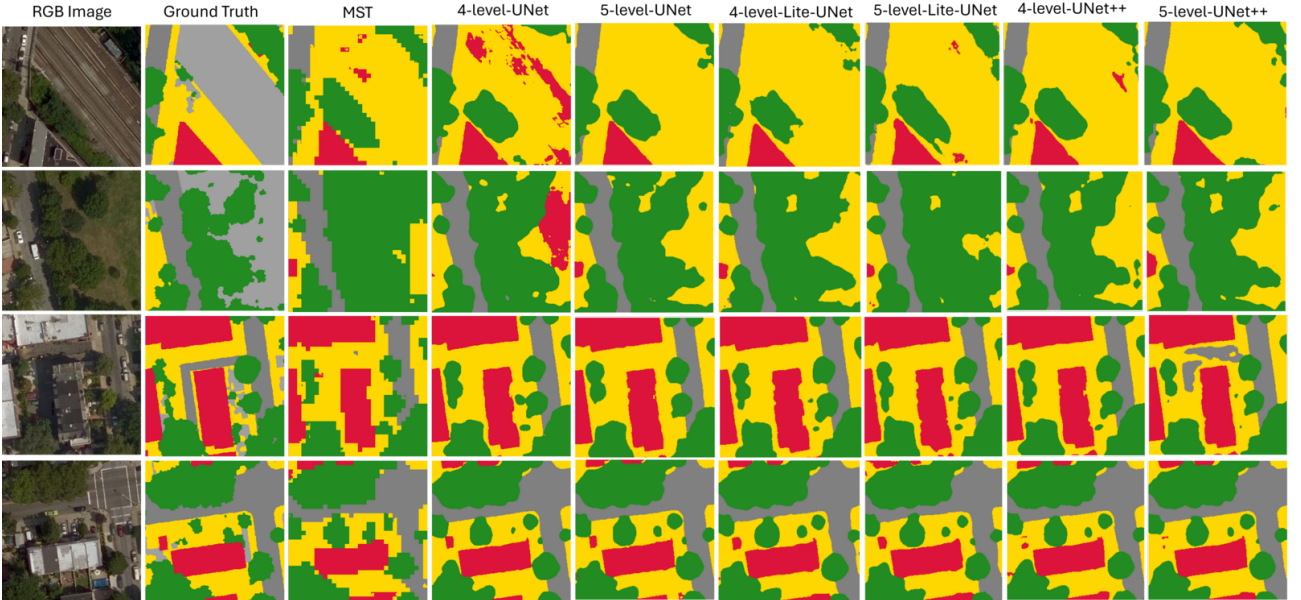


Figure 2: Qualitative comparison on Queens validation samples. Columns: RGB image, Ground Truth, MST, U-Net (L4), U-Net (L5), Lite U-Net (L4), Lite U-Net (L5), U-Net++ (L4), and U-Net++ (L5). Colors: Green (Tree Canopy), Red (Buildings), Grey (Roads), Yellow (Other Impervious).

5 Conclusion

This study establishes that aerial semantic segmentation, as a foundational component for cross-view geolocalisation, can be significantly improved through architectural isolation and the application of convolutional inductive biases. By benchmarking the Multi-Scale Transformer (MST) against various U-Net family models, we demonstrate that standard convolutional encoder-decoder structures provide a more robust basis for urban land-cover mapping. The 5-level U-Net achieved the highest overall fidelity with 74.17% mIoU, while the 4-level Lite U-Net emerged as a highly efficient alternative, delivering competitive accuracy with a significantly reduced parameter footprint.

These findings suggest that while transformers are effective for high-level spatial reasoning in CVGL pipelines, they are outperformed by convolutional models in the specific task of dense land-cover prediction. This study has several implications for the design of future localisation systems: (i) it demonstrates that high-fidelity masks of stable landmarks like buildings and roads can be achieved with relatively low computational overhead, and (ii) it confirms that increasing model depth yields diminishing returns for this specific urban dataset.

Future research will integrate these optimised convolutional segmentation backbones back into unified transformer-based CVGL pipelines. By providing high-fidelity semantic guidance, we aim to investigate the degree to which improved segmentation accuracy directly translates into reduced spatial ambiguity and enhanced localisation precision across diverse and challenging urban environments.

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